

A Source-Explicit Conditional Common-Weight All- m Type-II Estimate and the Complementary Signed High-Order Frontier

Audited full-programme snapshot: conditional Gate closures, Ford–Maynard endgame,
and exact open ledger

Research draft

August 26, 2026

Abstract

This note records an audited public-facing state of a parity-sensitive Type-II programme. Part I studies a canonical common-weight direct operator. A row is indexed by a prime $r \asymp R$, a shift k , and $m \asymp M$, with $m' = m + kr \asymp M$. The exact determinant identity

$$m' A_e(t) - m B_e(t) = 2k$$

leads, after a source-preserving Poisson–Bruhat transport, to a moving-row family $c_r(\xi) = B_\xi F_\xi(r/R)$. Under five explicit source/analytic pins stated below—a one-sided transport and absolute source ceiling, fixed-state exclusion, the outer-root energy pin, projective source routing, and the literal recombination estimate—band-limited prime participation yields the required $R^{-1/4+o(1)}$ contraction. The finite participation inequality, projective ratio-class algebra, exponent ledger, and normalization compilers are separately machine-checked; the displayed source/analytic pins are not silently promoted into Lean theorems.

Part II records the complementary Gate-1B frontier. The low/medium conductor range has a fixed-power margin. In the high-conductor H7 branch the source prime lies in the primitive conductor, and the short inducing-cofactor Möbius sign cancels exactly against the induced Gauss factor. Under the explicit H7 source-normalization pin, prime-character collapse and a hybrid Poisson formula recover the former fixed-power deficit and reduce the problem to a natural-scale dual determinant. The remaining H7 task is an arbitrary-log saving for a centred seven-defect determinant variance. H8, H9, the same- q Gram residual, the cross-modulus source residual, the zero-mode/source locks, and Full Type-II reassembly remain open.

Part III records the strongest currently justified Ford–Maynard endgame. The operating width is $u_0 = 1/6$, safely above the published central threshold 0.1663; the finite Gate-0/2 bank proves $\sigma \geq 49/300$ and hence $\Omega(n) \leq 6$ for the relevant rough class. After the earlier Gate-0 comparison inputs are frozen, Gate 2 is reduced to a conditional endgame whose load-bearing analytic antecedents are the full Ford–Maynard Type-II estimate at width $1/6$ and the source-exact rough N_2 input. These antecedents remain external/uninhabited, so the final prime-pair conclusion is not activated.

No claim of Full Type-II reassembly, a Hardy–Littlewood prime-pair asymptotic, or twin-prime infinitude is made.

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1 Introduction

A recurring obstruction in parity-breaking sieve arguments is that strong fixed-slice estimates need not survive coherent averaging over the true source family. The correct object is therefore the source-restricted family operator, not a collection of unrelated local norms.

This note isolates a canonical direct Type-II operator and derives a conditional all- m mean-square bound under explicit source and analytic pins. Four structural ingredients drive the proof:

- (i) exact source and determinant-2 row geometry;
- (ii) source-preserving Fourier/Poisson transport to a smooth moving- r family;
- (iii) band-limited prime participation, converting smoothness in r into a moving-family contraction;
- (iv) an exact projective crossed-convolution identity for the zero-curvature branch.

The downstream motivation is the Ford–Maynard prime-producing sieve framework, where Type I and Type II information for an actual non-negative sequence is used simultaneously [FM24]. Their work also shows that a substantial Type II range is genuinely necessary for non-trivial lower bounds. We do not perform the full downstream reassembly here.

1.1 External analytic input

The participation step needs a prime number theorem in intervals of length $R^{3/4}$. Huxley’s classical short-interval prime number theorem gives, uniformly for $x \asymp R$,

$$\pi(x + H) - \pi(x) = (1 + o(1)) \frac{H}{\log x} \quad (H \geq x^{7/12+\varepsilon}),$$

for every fixed $\varepsilon > 0$ [Hux72]. Since $3/4 > 7/12$, the plateaux used below are longer than the range required by Huxley’s theorem, so no modern strengthening is needed. We also use the classical Bernstein inequality

$$\|P'\|_\infty \leq N \|P\|_\infty$$

for trigonometric polynomials of degree at most N [Zyg02].

1.2 Scope

The theorem proved below concerns the canonical *common-weight* direct source. In the authoritative source, the physical smoothing datum is one weight W_D ; row dependence that appears later is derived deterministically from this common datum through Fourier, reciprocity and Poisson transforms. An abstract interface allowing arbitrary independent edge-dependent weights is strictly more general and belongs to an upstream source-adaptation problem. Likewise, a global packet census and Full Type II reassembly are not part of the theorem proved here. This distinction is also machine-checked in the formal development: the Gate-1A direct closure certificate contains no global packet-dictionary or edge-dependent-D2 field.

2 Parameters and notation

Let $X \rightarrow \infty$. Set

$$M = X^{1/3}, \quad R = X^a, \quad L = X^b,$$

and

$$H = X^{a+2b-2/3}, \quad K = \frac{M}{R}, \quad U = \frac{L}{H}, \quad D = \frac{L^2}{H}.$$

We repeatedly use

$$RK = M, \quad DH = L^2, \quad M^2H = RL^2. \quad (1)$$

The three controlling vertices are

$$V_1 = \left(\frac{5}{18}, \frac{1}{3} \right), \quad V_2 = \left(\frac{5}{18}, \frac{25}{72} \right), \quad V_3 = \left(\frac{7}{24}, \frac{1}{3} \right).$$

For a prime ℓ , define

$$\rho_\ell(n) := \mathbf{1}_{\ell|n} - \frac{1}{\ell}. \quad (2)$$

Let b_p, d_q be 1-bounded coefficient sequences on primes $p, q \asymp L$. All harmless divisor and logarithmic losses are absorbed in $X^{o(1)}$.

We use $e(z) = e^{2\pi iz}$ and $e_c(z) = e(z/c)$. For a Schwartz function f we use the Fourier convention

$$\widehat{f}(\xi) = \int_{\mathbb{R}} f(x) e(-x\xi) dx.$$

Every inverse symbol in a modular phase is understood only on the explicitly routed unit/full-conductor sector.

3 The canonical physical source

3.1 Rows and affine forms

A canonical row is $e = (r, k, m)$ with

$$r \asymp R \text{ prime}, \quad 1 \leq |k| \leq K, \quad m \asymp M, \quad m' := m + kr \asymp M,$$

together with the generic unit/full-conductor conditions needed by the reciprocal phases. Let $\mathcal{E}^\sharp$ be this row family.

Choose w_0 modulo r such that

$$mw_0 \equiv -2 \pmod{r}.$$

Since $m' = m + kr \equiv m \pmod{r}$, the same choice satisfies $m'w_0 \equiv -2 \pmod{r}$. Hence

$$\alpha := \frac{mw_0 + 2}{r}, \quad \beta := \frac{m'w_0 + 2}{r}$$

are integers. Define

$$A_e(t) := \alpha + mt, \quad B_e(t) := \beta + m't.$$

Lemma 3.1 (Determinant-2 row identity). *For every canonical row and every integer t ,*

$$m'A_e(t) - mB_e(t) = 2k.$$

Proof. The t -terms cancel and

$$m'\alpha - m\beta = \frac{m'(mw_0 + 2) - m(m'w_0 + 2)}{r} = \frac{2(m' - m)}{r} = 2k.$$

□

3.2 Common physical smoothing

Fix $W \in C_c^\infty(\mathbb{R})$ and write $W_D(t) = W(t/D)$. The key point is that this weight is common to all rows. Define

$$P_e(t) = \sum_{p \asymp L} b_p \rho_p(A_e(t)), \quad Q_e(t) = \sum_{q \asymp L} d_q \rho_q(B_e(t)),$$

and

$$C_e(b, d) := \sum_{t \in \mathbb{Z}} W_D(t) P_e(t) Q_e(t). \quad (3)$$

Remark 3.2. An arbitrary family $W_{D,e}$ is not part of the canonical theorem. Such weights arise only in a more general source-adaptation interface.

3.3 Source provenance and derived row dependence

The common-weight assertion is stronger than a notational convention. After Fourier transformation the source coefficient factors as

$$\omega_{m,p,q}(h) = \underbrace{\frac{H}{pq} e\left(-\frac{h\alpha}{pqm}\right)}_{\text{derived row-dependent scalar}} \underbrace{\widehat{W}_D\left(\frac{h}{pq}\right)}_{\text{one common physical weight}}, \quad (4)$$

where the argument $h/(pq)$ has no row index. Thus the later dependence on m, r, p, q is *derived dependence*; it is not freedom to choose an independent physical weight on each row. This distinction is load-bearing: an arbitrary edge-dependent family would be a strictly stronger source class and is deliberately excluded from the canonical direct theorem.

4 Fourier normalization and determinant shell

The normalized pre-square source is

$$\tilde{C}_{r,k,m} = \sum_{p \asymp L} b_p \sum_{\substack{q \asymp L \\ q \neq p}} d_q \sum_h \omega_{m,p,q}(h) e_{mq}\left(2hk\overline{p(m+kr)}\right), \quad (5)$$

where

$$\omega_{m,p,q}(h) = \frac{H}{pq} \widehat{W}_D\left(\frac{h}{pq}\right) e\left(-\frac{h\alpha}{pqm}\right). \quad (6)$$

The source norms satisfy

$$\sum_h |\omega_{m,p,q}(h)|^2 \ll H, \quad \sum_h |\omega_{m,p,q}(h)| \ll H.$$

Writing $m' = m + rk$, additive reciprocity separates the phase:

$$e_{mq}(2hk\overline{pm'}) = e_{pr}(-2h\overline{qm}) e\left(\frac{2h}{prqm}\right) e_q(-2h\overline{prmm'}). \quad (7)$$

Thus the exact H -scale smooth profile is

$$\beta_{m,r;p,q}(h/H) = \frac{H}{pq} \widehat{W}_D(h/pq) e\left(-\frac{h\alpha}{pqm} + \frac{2h}{prqm}\right).$$

The h -Poisson/Gauss completion gives the determinant shell

$$ms - prn = 2. \quad (8)$$

Choose $\kappa_p(m) \in [0, pr)$ with $m\kappa_p(m) \equiv 2 \pmod{pr}$ and set

$$n_0(m, p, r) = \frac{m\kappa_p(m) - 2}{pr}.$$

Then every solution is parameterized by

$$s_j = \kappa_p(m) + jpr, \quad n_j = n_0 + jm.$$

The quotient amplitude is

$$\mathcal{W}_{m,r;p,q}(j) := \widehat{\beta}_{m,r;p,q} \left(\frac{H(n_0 + jm)}{qm} \right), \quad (9)$$

and, for $|j| \ll U = L/H$,

$$\sum_{|j| \ll U} |\mathcal{W}_{m,r;p,q}(j)|^2 \ll UX^{o(1)}. \quad (10)$$

5 Source-preserving Poisson–Bruhat transport

The following source-specific statement is isolated because it is the exact bridge between the authoritative physical source and the finite BPP compiler. The scale identity in Lemma 5.3 is elementary; the uniform smooth envelope, the negligible-state disposal, and the numerical source ceiling are source/analytic inputs.

For a transported state family put

$$a_{r,\xi} := |c_r(\xi)|, \quad \mathcal{T}_{\text{abs}} := \sum_{\substack{r \in [R, 2R] \\ r \text{ prime}}} \left(\sum_{\xi} a_{r,\xi} \right)^2.$$

Hypothesis 5.1 (One-sided source transport and absolute ceiling). *For every fixed $A_0 > 0$, the transported source splits into a retained finite state set $\mathfrak{X}_{\text{ret}}$ and a discarded remainder with Gate-energy contribution $O_{A_0}(X^{-A_0})$ times the natural scale. Every retained state has scalar moving-row amplitude*

$$c_r(\xi) = B_{\xi} F_{\xi}(r/R), \quad (11)$$

where B_{ξ} is independent of r , F_{ξ} is supported in a fixed compact subinterval of $(1, 2)$, and for every fixed J ,

$$\sup_{\xi} \sup_t |\partial_t^J F_{\xi}(t)| \ll_J X^{o(1)}.$$

For every fixed truncation exponent A , the retained set is chosen so that $\|F_{\xi}\|_{\infty} \geq 10R^{-A}$; states failing this inequality are part of the discarded source remainder. The transport is one-sided, has $X^{o(1)}$ nuclear cost, and gives the explicit absolute source ceiling

$$\boxed{\mathcal{T}_{\text{abs}} \ll M^2 L^4 X^{o(1)}}. \quad (12)$$

No two-sided Hilbert–Schmidt equivalence is assumed. Finally, the physical and normalized sources obey

$$C_e = H^{-1} \widetilde{C}_e + \mathcal{R}_e, \quad \sum_{e \in \mathcal{E}^{\sharp}} |\mathcal{R}_e|^2 \ll_{A_0} X^{-A_0} \frac{ML^4}{H},$$

for every fixed $A_0 > 0$.

Status note 5.2 (Source status). The finite Poisson–Bruhat algebra and the budget identity below are banked. The full assertion in Hypothesis 5.1, including the literal source equality, uniform retained-state truncation, the physical-normalized bridge and (12), is an explicit source/analytic pin. It is not inferred from the exponent arithmetic.

Lemma 5.3 (Poisson–Bruhat budget identity). *The scale identities imply*

$$\boxed{\frac{MK(MH)^2}{RL^4} = 1.}$$

Proof. By (1), $MK = L^2/H$. Therefore

$$\frac{MK(MH)^2}{RL^4} = \frac{(L^2/H)M^2H^2}{RL^4} = \frac{M^2H}{RL^2} = 1.$$

□

Remark 5.4. The amplitude is smooth in r but is not constant. Thus a count of the prime divisors of a fixed determinant does not by itself give an R^{-1} weighted family saving.

6 Band-limited prime participation

From now on ξ ranges over the retained state set of Hypothesis 5.1. Put $a_{r,\xi} = 0$ when the prime row r is excluded for the state ξ , and define

$$M_\xi := |B_\xi| \sup_{1 \leq t \leq 2} |F_\xi(t)|.$$

Fix a large A and put $N = R^{1/4}$. A smooth periodic extension of F_ξ and repeated integration by parts give, uniformly in retained ξ ,

$$F_\xi(t) = P_\xi(t) + O_A(R^{-A}), \tag{13}$$

where P_ξ is a trigonometric polynomial of degree at most N .

Lemma 6.1 (Bernstein plateau). *For every retained state ξ , there is an interval $I_\xi \subset [R, 2R]$ of length $\gg R^{3/4}$ on which $|c_r(\xi)| \gg M_\xi$.*

Proof. Let t_ξ maximize $|P_\xi|$. Bernstein’s inequality [Zyg02] gives $\|P'_\xi\|_\infty \leq N\|P_\xi\|_\infty$. Hence on a one-sided or two-sided interval of t -length $\gg N^{-1}$ contained in $[1, 2]$, P_ξ retains a fixed proportion of its maximum. The retained-state clause in Hypothesis 5.1 makes the error in (13) smaller than that maximum. Multiplying the interval length by R gives $R/N = R^{3/4}$. □

Hypothesis 6.2 (Fixed-state exclusion). *For each retained state ξ , every prime row excluded from the generic non-projective branch divides one of $X^{o(1)}$ fixed nonzero integers $N_{\xi,j}$ of polynomial height, $|N_{\xi,j}| \leq X^{O(1)}$.*

Lemma 6.3 (Admissible prime participation). *Assume Hypothesis 6.2. For every retained state ξ , the plateau I_ξ contains $R^{3/4-o(1)}$ admissible prime rows. In particular, if*

$$A_\xi := \max_{\substack{r \sim R \\ r \text{ prime}}} a_{r,\xi}, \quad N_\xi := \frac{\sum_{r \sim R, r \text{ prime}} a_{r,\xi}}{A_\xi},$$

then $A_\xi > 0$ and

$$A_\xi \asymp M_\xi, \quad N_\xi \gg R^{3/4-o(1)}. \tag{14}$$

Proof. Huxley's short-interval prime number theorem applies to the interval from Lemma 6.1, since $3/4 > 7/12 + \varepsilon$ for a fixed small $\varepsilon > 0$ [Hux72]. Thus I_ξ contains $R^{3/4-o(1)}$ primes. A nonzero polynomial-height integer has $O(\log X)$ distinct prime divisors, and only $X^{o(1)}$ such integers occur, so the exclusion removes only $X^{o(1)}$ primes. On every remaining plateau prime, $a_{r,\xi} \gg M_\xi$, proving both claims. $\square$

Let $\mathcal{R}$ be the primes in $[R, 2R]$, let $S = |\mathcal{R}| = R^{1+o(1)}$, and retain the notation $\mathcal{T}_{\text{abs}}$ from Section 5.

Lemma 6.4 (Participation-to-envelope). *If $N_\xi \geq P$ for every retained state, then*

$$\left(\sum_{\xi} A_{\xi} \right)^2 \leq \frac{S}{P^2} \mathcal{T}_{\text{abs}}.$$

Proof. By Cauchy–Schwarz,

$$\left(\sum_{\xi} A_{\xi} N_{\xi} \right)^2 = \left(\sum_{r \in \mathcal{R}} \sum_{\xi} a_{r,\xi} \right)^2 \leq S \mathcal{T}_{\text{abs}}.$$

Since $N_{\xi} \geq P$, the result follows. $\square$

6.1 Projective versus non-projective Gram pairs

The transport attaches to every retained state a pair (Z_{ξ}, L_{ξ}) and the outer determinant

$$\Delta(\xi, \eta) := Z_{\xi} L_{\eta} - Z_{\eta} L_{\xi}. \quad (15)$$

We split the Gram pair set before applying BPP:

$$\Delta(\xi, \eta) \neq 0 \quad \text{or} \quad \Delta(\xi, \eta) = 0.$$

The second class is the projective branch in Section 8. Since $\Delta(\xi, \xi) = 0$, the entire state diagonal lies in that branch.

For $\Delta(\xi, \eta) \neq 0$, a row prime can couple the fixed pair only if it divides the fixed nonzero polynomial-height integer $\Delta(\xi, \eta)$. Hence

$$\mathcal{E}_{\text{np}} \ll X^{o(1)} \left(\sum_{\xi} A_{\xi} \right)^2. \quad (16)$$

Proposition 6.5 (BPP family-energy contraction). *Under Hypotheses 5.1 and 6.2,*

$$\boxed{\mathcal{E}_{\text{np}} \ll R^{-1/2+o(1)} \mathcal{T}_{\text{abs}}.}$$

In particular, by (12),

$$\mathcal{E}_{\text{np}} \ll R^{-1/2+o(1)} M^2 L^4.$$

Proof. Use Lemma 6.4 with $P = R^{3/4-o(1)}$ and $S = R^{1+o(1)}$ in (16). $\square$

7 The unique outer square root

The finite outer four-cycle inequality gives a single square root, but its application to the canonical Gate energy requires an exact energy pin. We state that pin explicitly.

Hypothesis 7.1 (Outer-root energy pin). *Let $\mathcal{G}_{\text{np}}$ denote the normalized Gate contribution of the non-projective transported branch. The source identification and the finite outer-four-cycle compiler give*

$$\mathcal{G}_{\text{np}} \ll M^2 L^4 X^{o(1)} \left(\frac{\mathcal{E}_{\text{np}}}{\mathcal{T}_{\text{abs}}} \right)^{1/2}. \quad (17)$$

The finite square-root algebra behind (17) is

$$\sum_r \|A_r\|_{\text{op}} \leq |\mathcal{R}|^{1/2} \left(\sum_r \mathcal{T}_r \right)^{1/2},$$

and is machine-checked. The equality identifying its abstract energy with the canonical Gate energy is the source pin in Hypothesis 7.1. Combining Proposition 6.5 and Hypothesis 7.1 gives

$$\mathcal{G}_{\text{np}} \ll R^{-1/4+o(1)} M^2 L^4.$$

The required comparison is $MR^{-1/4} \leq H$, equivalently

$$\frac{5}{4}a + 2b - 1 \geq 0.$$

At V_1, V_2, V_3 the margins are respectively

$$\frac{1}{72}, \quad \frac{1}{24}, \quad \frac{1}{32}.$$

8 Projective reassembly: ratio classes

The complementary branch is $\Delta(\xi, \eta) = 0$, including $\xi = \eta$. On $L \neq 0$ the correct class is the projective ratio $\kappa(Z, L) = Z/L$, not the product ZL . For a finite transported state support $\mathcal{S}$, put $v_{Z,L} = A_Z \otimes B_L$ and

$$C_\xi := \sum_{(Z,L) \in \mathcal{S}: \kappa(Z,L)=\xi} v_{Z,L}.$$

Then the exact class-pushforward identity is

$$\sum_{\xi} \|C_\xi\|^2 = \sum_{(Z_1, L_1), (Z_2, L_2) \in \mathcal{S}} \mathbf{1}_{Z_1 L_2 = Z_2 L_1} \text{Re} \langle v_{Z_1, L_1}, v_{Z_2, L_2} \rangle. \quad (18)$$

Hypothesis 8.1 (Projective source routing). *The transported $\Delta = 0$ source is routed into positive primitive ratio-class packets of class multiplicity $X^{o(1)}$. One-zero axes and non-primitive/rank-loss states are routed to explicit exceptional blocks. Moreover the diagonal/projective source mass satisfies*

$$\sum_{(Z,L) \in \mathcal{S}} \|A_Z\|^2 \|B_L\|^2 \ll MHL^4 X^{o(1)}, \quad (19)$$

and every exceptional block is bounded at the same target.

Under Hypothesis 8.1, Cauchy within each ratio class and (18) give

$$\mathcal{G}_{\text{proj}} \ll MHL^4 X^{o(1)}.$$

The finite ratio-class identity, primitive rigidity and multiplicity lemma are machine-checked; Hypothesis 8.1 is the physical source-routing pin.

9 Recombination error

Hypothesis 9.1 (Literal source recombination estimate). *The quotient-cell/source recombination error obeys*

$$\mathcal{E}_{\text{rec}} \ll U^{-2+o(1)} M^2 L^4. \quad (20)$$

Since $U = L/H$ and $DH = L^2$,

$$\frac{U^{-2} M^2 L^4}{MHL^4} = \frac{M}{D}.$$

The margins at V_1, V_2, V_3 are $1/18, 1/18, 1/24$.

10 Conditional main theorem

Theorem 10.1 (Canonical all- m common-weight estimate). *Let the parameters lie in the polytope generated by V_1, V_2, V_3 . Let $\mathcal{E}^\sharp$ be the generic unit/full-conductor all- m row family and let $C_e(b, d)$ be the covariance (3). Assume Hypotheses 5.1, 6.2, 7.1, 8.1 and 9.1. Then*

$$\sum_{e \in \mathcal{E}^\sharp} |C_e(b, d)|^2 \ll \frac{ML^4}{H} X^{o(1)}.$$

Equivalently,

$$\sum_{e \in \mathcal{E}^\sharp} |\tilde{C}_e(b, d)|^2 \ll MHL^4 X^{o(1)}.$$

Proof. The non-projective branch is bounded by Proposition 6.5 and Hypothesis 7.1. The complete $\Delta = 0$ branch, including the state diagonal, is bounded by Hypothesis 8.1 and the exact ratio-class identity. The discarded source and the recombination error are controlled by Hypotheses 5.1 and 9.1. These branches are exhaustive and disjoint. The normalized estimate follows. The physical-normalized identity in Hypothesis 5.1, together with its square-summable remainder, gives the physical estimate. $\square$

Corollary 10.2 (Clean- P_3 restriction). *Every clean- P_3 subfamily of $\mathcal{E}^\sharp$ satisfies the same bound.*

Proof. The energy is a sum of non-negative row energies. $\square$

11 Scope separation: Gate 1A versus source routing

The theorem above is a conditional analytic theorem for the canonical direct operator. It does not assert that every packet in a global sieve decomposition has already been identified with this operator.

Suppose an upstream source compiler gives an exact decomposition

$$\mathcal{S}_{\text{actual}} = \sum_{\lambda \in \Lambda_{\text{gen}}} a_\lambda D_\lambda U_\lambda \mathcal{S}_\lambda^\sharp + \mathcal{S}_{\text{exc}},$$

where U_λ is unitary/permutational and

$$\sum_{\lambda} |a_\lambda| \|D_\lambda\|_{\text{op}} \leq X^{o(1)}.$$

Then the main theorem controls the generic packets by Minkowski, provided each packet is truly an instance of the canonical operator. Proving the source identity, packet multiplicity, and exceptional routing is a separate Gate-0 compiler problem.

In particular, a common physical weight routes directly into the theorem. A fixed-dimensional smooth family generated deterministically from the common source can be decomposed into $X^{o(1)}$ common templates. By contrast, an arbitrary independently varying edge weight is a genuinely stronger object. In the formal development this more general datum is retained as an upstream Gate-0 source-adaptor obligation; it is not a premise of the canonical Gate-1A direct theorem.

This separation matters downstream. The Ford–Maynard framework applies to the actual non-negative sequence and its complete Type-I/Type-II information [FM24]; a convenient analytic model cannot replace an unverified source census.

12 Formal verification and provenance

The accompanying Lean bank deliberately separates finite mathematics, source provenance, and deep analytic input. This is useful here because a theorem compiler can be perfectly correct while its external analytic hypothesis is not itself formalized in Mathlib.

12.1 Machine-checked finite/source skeleton

The v9.8 formal bank verifies, without user axioms or admitted proofs:

- the canonical pre-square direct source as a literal source transcription, together with the fact that the physical transform $\widehat{W}_D$ is one common datum;
- the all- m generic row type, with no clean- P_3 field;
- the exact normalized and physical Gate-1A energies and their H^2 normalization bridge;
- the finite participation-to-family-energy inequality;
- the exponent conversion $R^{-1/2} \mapsto R^{-1/4}$ and the margins $1/72, 1/24, 1/32$;
- the projective/axis finite algebra and the U^{-2} exponent budget;
- the logical separation between the Gate-1A direct theorem and the upstream global packet/source census.

The relevant files are

<code>V98CanonicalDirectSource.lean</code>	canonical source and common-weight firewall
<code>V98CanonicalAllMRows.lean</code>	all- m row family
<code>V98DirectEnergyPin.lean</code>	exact energy objects and pinning interface
<code>V98BPPProvenance.lean</code>	external BPP participation provenance
<code>V98DirectClosure.lean</code>	deterministic normalized-to-physical compiler
<code>V98Gate0ScopeSplit.lean</code>	Gate-0/Gate-1A scope separation
<code>V98Status.lean</code>	axiom/status audit

12.2 What Lean does not prove

The formal development does *not* manufacture an inhabitant of the deep analytic BPP source input. In particular, the trigonometric Bernstein estimate, the source-specific smooth truncation, and Huxley’s short-interval prime theorem are not currently formalized in the

project. They are recorded with external/source-specific provenance, and no global axiom is introduced. The state-diagonal issue is not external: the finite moving-family bank splits diagonal and off-diagonal exactly, and the corrected projective identity places the diagonal in the $\Delta = 0$ branch.

Likewise, the formal “energy pin” is an explicit equality/interface tying a generic BPP energy to the canonical Gate-1A energy; the development includes guards showing that an arbitrary abstract bound cannot manufacture this pin. Thus a successful compiler is not confused with an independently proved analytic estimate.

12.3 Research status versus formal status

Accordingly the appropriate status language is:

source-explicit conditional theorem under the displayed analytic/source pins

for the mathematics of this note, and

external-uninhabited in Lean

for the fully formal end-to-end analytic proof. The latter is a provenance boundary, not an exponent deficit or an additional arithmetic obstruction.

13 Relation to Kloosterman-fraction technology

Bettin–Chandee prove non-trivial bounds for separated trilinear Kloosterman fractions with arbitrary coefficient sequences [BC18]. Wright improves this architecture when a factor of the denominator is fixed [Wri26]. These results motivated intermediate representations, but the final direct proof here does not black-box either theorem. Its decisive resource is instead smooth participation in the moving prime family.

14 What remains outside this theorem

The following are not conclusions of the main theorem:

1. exhaustive routing of every global high-complexity source packet into the canonical direct operator;
2. closure of the separate Gate-1B analytic frontier, whose present high-conductor residual is recorded in Section C;
3. Full Type-II reassembly;
4. a Hardy–Littlewood prime-pair asymptotic or twin-prime infinitude.

The point of the second item is structural. The conditional Gate-1A theorem established here supplies one all- m analytic engine once its displayed pins are verified. It does not provide the signed prime-conductor character moment that remains in the complementary Gate-1B branch, and hence cannot by itself be promoted to a complete Type-II theorem.

15 Conclusion

The main conceptual point is that local sparsity alone does not give a moving-family saving. After source-preserving transport, the row coefficient remains r -dependent through a uniformly smooth envelope. Band-limited prime participation supplies the missing family coherence: smoothness creates a long plateau, prime distribution fills it with many moving rows, and finite Cauchy converts participation into the required $R^{-1/2+o(1)}$ energy contraction. After one and only one outer square root, the resulting $R^{-1/4+o(1)}$ gain closes the canonical all- m common-weight Gate-1A direct estimate throughout the parameter polytope.

A substantive feature of this mechanism is that its resource is the moving prime row r , not a prime factor of m . No largest-prime-factor quantity of m enters Lemmas 6.1 and 6.3 and Proposition 6.5. Thus the argument applies uniformly to the canonical all- m row family, including smooth values of m ; clean- P_3 is a positive-restriction corollary rather than the source of the saving.

A Formal theorem map and scope ledger

Component	Status	Role in this note
Canonical source and common W_D	source-pinned; finite identities checked	fixes the source class
All- m row family	Lean-constructed	removes clean- P_3 from the statement
PB smooth envelope, retained-state disposal and $\mathcal{T}_{\text{abs}}$ ceiling	conditional source/pin	supplies the absolute comparator
Finite BPP compiler	Lean-proved	participation $\Rightarrow$ family energy
Huxley + fixed-state exclusion	published theorem + source pin	supplies prime participation
Outer-root energy pin	conditional source pin ; square-root algebra proved	identifies the abstract energy with the Gate energy
Ratio-class projective algebra	Lean-proved finite algebra	includes the diagonal in $\Delta = 0$
Physical projective routing and source ceiling	conditional source pin	bounds the zero-curvature branch
U^{-2} exponent budget	Lean-proved arithmetic	checks the recombination margin
Literal recombination estimate	conditional source pin	controls the source error
Arbitrary edge-dependent packet	outside theorem scope	upstream adapter obligation
Global packet census / Full Type II	open	not implied here

B Canonical source versus arbitrary edge dependence

It is useful to make the source distinction algebraically explicit. The canonical coefficient has the form

$$\omega_e(p, q, h) = R_e(p, q, h) \widehat{W}_D(h/(pq)),$$

where R_e is a deterministic scalar generated by the row and by exact reciprocity. Hence the entire family is determined once the one physical weight W_D is fixed. This is fundamentally different from prescribing an arbitrary collection

$$\{W_{D,e} : e \in \mathcal{E}^\# \}$$

with no relation between different rows. The latter can have effective rank comparable with the number of rows and therefore cannot be reduced to the canonical theorem by functional analysis alone. This is why the global source compiler must either identify a packet with the common-weight source, reduce it to finitely many smooth templates, or route it elsewhere.

C Complementary programme boundary: Gate 1B

This appendix is a research-status calculation, not an ingredient in Theorem 10.1. “Exact” below refers to the displayed squarefree/unit source dictionary and Fourier convention. The natural-scale estimate in Proposition C.5 additionally uses the explicit source norm in Hypothesis C.4.

C.1 Low/medium conductor and the high-conductor threshold

Write $q = ce$, $c \asymp C$, $q \asymp Q$. After the corrected state count,

$$\sum_{c, \chi^*, e} |\beta(ce)|^2 \ll QC X^{o(1)}.$$

The coefficient-blind estimate has the relative shape

$$\frac{|\mathcal{H}_s(C)|}{Y^9} \ll \begin{cases} C/Q, & C \leq Y^4, \\ C^2/(QY^4), & C \geq Y^4. \end{cases}$$

The formulae cross at $C = Y^4$. The large- C estimate loses all saving at

$$C_0 = Q^{1/2}Y^2,$$

so C_0 is the zero-saving/high-conductor threshold, not the crossing point.

C.2 Exact source split and induced-Gauss cancellation

The squarefree coefficient is

$$\beta(q) = \sum_{dp=q} \mu(d) \log p,$$

with $p > V$ and $q/p > U$. In high conductor, $e = q/c \leq Y^2$, whereas $V = Y^{5/2-o(1)}$, hence $p > V > e$ and the lane $p \mid e$ is empty. Thus

$$c = pc_0, \quad d = c_0e,$$

and

$$\beta(ce) = \mu(e)\rho_{C,e}(c), \quad \rho_{C,e}(c) = \sum_{pc_0=c}^{\text{physical}} \mu(c_0) \log p. \quad (21)$$

For a primitive $\chi^* \pmod{c}$ induced to ce ,

$$\tau_{ce}(\chi) = \mu(e)\chi^*(e)\tau_c(\chi^*). \quad (22)$$

Consequently the short-cofactor sign is spent exactly:

$$\beta(ce)\tau_{ce}(\chi) = \rho_{C,e}(c)\chi^*(e)\tau_c(\chi^*).$$

It cannot be used a second time as cancellation.

C.3 Prime-character collapse

Write $c = pc_0$, $(p, c_0) = 1$, and factor a primitive character as $\chi = \chi_p \chi_0$. Let $N = r_1 \cdots r_7 x \asymp Y^8$. The p -local argument is

$$A_p \equiv c_0 eN \overline{2h} \pmod{p}.$$

For $A \in \mathbb{F}_p^\times$,

Lemma C.1 (Prime-character collapse).

$$\sum_{\chi_p \neq \chi_{0,p}} \frac{\tau_p(\chi_p)}{\sqrt{p}} \chi_p(A) = \frac{p-1}{\sqrt{p}} e_p(\bar{A}) + \frac{1}{\sqrt{p}}.$$

Proof. Summing first over all multiplicative characters gives $(p-1)e_p(\bar{A})$. The principal character has Gauss sum -1 , so removing it adds 1. $\square$

The main term therefore becomes the linear additive phase

$$e_p(2h \overline{c_0 eN}).$$

The correction is smaller than the main local multiplier by p^{-1} ; since $p > V = Y^{5/2-o(1)}$, it is power-negligible under the same absolute source ceiling used below.

C.4 Hybrid Poisson formula and primitive projector

Use $\widehat{v}(\xi) = \int v(x) e(-x\xi) dx$.

Lemma C.2 (Hybrid Poisson). *Let $p \nmid c_0$, and let χ_0 be primitive modulo c_0 . With the consistent Fourier sign convention,*

$$\sum_{h \in \mathbb{Z}} v(h/H) \overline{\chi_0(h)} e_p(ah) = \frac{H}{c_0} \tau_{c_0}(\overline{\chi_0}) \sum_{\substack{m \in \mathbb{Z} \\ m \equiv a c_0 \pmod{p}}} \chi_0(-m\bar{p}) \widehat{v}\left(\frac{mH}{pc_0}\right).$$

Proof. Apply Poisson summation in residue classes modulo pc_0 . The finite Fourier sum factors by CRT. The p -factor enforces the displayed congruence, while the c_0 -factor is the twisted Gauss sum $\tau_{c_0}(\overline{\chi_0}) \chi_0(-m\bar{p})$. The factor p from the p -sum cancels the p in the Poisson denominator. $\square$

Here $a = 2\overline{c_0 eN} \pmod{p}$, so the surviving frequency obeys

$$meN \equiv 2 \pmod{p}, \quad |m| \ll (Y/e) X^{o(1)} < p.$$

For odd squarefree c_0 and $(A, c_0) = 1$,

Lemma C.3 (Primitive-character projector).

$$\sum_{\chi_0 \pmod{c_0}}^* \chi_0(A) = \sum_{\substack{d|c_0 \\ d|A-1}} \mu(c_0/d) \phi(d).$$

Consequently

$$\frac{\mu(c_0)}{c_0} \sum_{\chi_0}^* \chi_0(A) = \frac{1}{c_0} \sum_{\substack{d|c_0 \\ d|A-1}} \mu(d) \phi(d).$$

Proof. This is Möbius inversion on the conductor of a character. The second identity uses squarefreeness: $\mu(c_0)\mu(c_0/d) = \mu(d)$. $\square$

With $A = meN/2$, the two projectors combine into $pd \mid meN - 2$.

C.5 Dual determinant and the fixed-power repair

The main term is reduced to

$$\begin{aligned} \mathcal{D}_7(C) = & \sum_{\substack{pc_0 \asymp C \\ p > V \\ e \asymp Q/C}} \log p \left(1 - \frac{1}{p}\right) \sum_{N \asymp Y^8} a_7(N) \sum_{\substack{|m| \ll (Y/e)X^{o(1)} \\ p | meN-2}} \widehat{v}(me/Y) \\ & \times \frac{1}{c_0} \sum_{\substack{d | c_0 \\ d | meN-2}} \mu(d) \phi(d). \end{aligned} \quad (23)$$

Hypothesis C.4 (H7 source norm and cell multiplicity). *The literal seven-defect plus one-model coefficient satisfies*

$$\sum_{N \asymp Y^8} |a_7(N)| \ll Y^8 X^{o(1)},$$

$\widehat{v}$ is rapidly decreasing with bounded supremum, and the omitted physical dyadic/coprimality selectors have $X^{o(1)}$ total multiplicity. The primitive character family in Lemma C.3 is the literal family produced by the source dictionary (with any parity restriction separately accounted for).

Proposition C.5 (Natural-scale fixed-power repair). *Under Hypothesis C.4,*

$$|\mathcal{D}_7(C)| \ll Y^9 X^{o(1)}.$$

Proof. For fixed (e, m, N) , the nonzero integer $meN - 2$ has $O(1)$ prime divisors $p > V = Y^{5/2-o(1)}$. After p is fixed, write $c_0 = ds$. The absolute divisor projector is bounded by

$$\sum_{d | meN-2} \sum_{s \asymp C/(pd)} \frac{\phi(d)}{ds} \ll X^{o(1)},$$

because the s -sum is harmonic on a dyadic interval and the divisor count is $X^{o(1)}$. Summing $e \asymp E$ and $|m| \ll Y/e$ costs $O(YX^{o(1)})$. Finally use the ℓ^1 source norm of a_7 . $\square$

Thus the former coefficient-blind fixed-power loss Q/Y^4 is recovered, but the target is $Y^9(\log X)^{-A}$. The remaining problem is therefore the arbitrary-log estimate

$$\boxed{\mathcal{D}_7(C) \ll_A Y^9 (\log X)^{-A}.}$$

Equivalently, one may formulate a centred seven-defect determinant variance for

$$nN - pd\ell' = 2, \quad n \ll Y, \quad N \asymp Y^8.$$

This estimate is open. The transform also has an anti-loop: the full divisor term $d = c_0$ reconstructs the original two-model determinant skeleton, so the Poisson/character transform is not an iteration-to-closure by itself.

C.6 Remaining Gate-1B ledger

Even a proof of the H7 logarithmic estimate would not close Gate 1B. The remaining items include H8, H9, the same- q nine-factor Gram, the underlying cross-modulus residual, literal source-normalization pins, the canonical zero-mode lock, global packet census and Full Type-II reassembly.

D Gate 0, Gate 2, and the Ford–Maynard endgame

The preceding sections isolate the two Type-II analytic engines. To make the programme as close to a complete conditional theorem as the present evidence permits, we now state the remaining Gate-0/Gate-2 interface explicitly. This section is downstream bookkeeping and source analysis; it is not used in the proof of the conditional Gate-1A theorem.

D.1 Gate 0: deterministic Type-I compiler and research status

The separate Gate-0 bank works with a shifted-prime sequence a_n , a simpler comparison sequence b_n , and $w_n = a_n - b_n$. Its deterministic compiler has the following logical form. A suitable residue-class weighted maximal Bombieri–Vinogradov input for a_n , together with the comparison progression mean for b_n and the exact reindexing bridge, implies the Ford–Maynard Type-I predicate for w_n . The finite compiler is machine-checked, while the deep analytic inputs remain external to Lean.

Accordingly we record the Gate-0 status as

research-level analytic pass; external-uninhabited in Lean.

This status does not contain any Type-II information and therefore cannot by itself activate the Ford–Maynard lower-bound theorem.

D.2 The operating Ford–Maynard parameters

For the downstream application we choose the central operating width

$$\nu_0 = \frac{1}{6}.$$

For $\varepsilon > 0$ set

$$P_\varepsilon = \left(\frac{1}{2} - \varepsilon, \varepsilon, \frac{1}{6} - 2\varepsilon \right). \quad (24)$$

The corresponding Type-II interval in the Ford–Maynard convention is

$$[\theta, \theta + \nu] = \left[\varepsilon, \frac{1}{6} - \varepsilon \right]. \quad (25)$$

The published central threshold in Ford–Maynard Theorem 2.7 is 0.1663; our operating value is safely above it, irrespective of the harmless strict/non-strict convention at the printed endpoint:

$$\frac{1}{6} - 0.1663 = \frac{11}{30000} > 0. \quad (26)$$

Ford–Maynard also use precisely the shrunk family $(1/2 - \varepsilon, \varepsilon, \nu - 2\varepsilon)$ in Theorem 2.7 [FM24]. Thus the appearance of the third coordinate $\nu - 2\varepsilon$ is part of the published parameter geometry rather than a new sieve hypothesis introduced here.

D.3 A finite roughness consequence

Write

$$\sigma = \nu_0 - 2\varepsilon.$$

For $0 \leq \varepsilon \leq \nu_0/100$, the Gate-0/2 finite bank proves

$$\sigma \geq \frac{49}{300}. \quad (27)$$

The following elementary lemma is fully independent of the deeper Ford–Maynard analysis.

Lemma D.1 (Rough numbers have at most six prime factors). *Let $n > 1$ and suppose every prime divisor of n is at least n^σ , where $\sigma \geq 49/300$. Then*

$$\Omega(n) \leq 6,$$

where prime factors are counted with multiplicity.

Proof. If $k = \Omega(n)$, then

$$n = \prod_{j=1}^k p_j \geq n^{k\sigma}.$$

Hence $k\sigma \leq 1$, so

$$k \leq \frac{1}{\sigma} \leq \frac{300}{49} < 7.$$

Since k is an integer, $k \leq 6$. □

This finite statement is machine-checked in the Gate-0/2 bank. It is useful because the final rough N_2 contribution lies in a bounded-complexity class once the required Ford–Maynard roughness identity has been supplied.

D.4 The rough-weight input and the N_2 cell sum

The remaining source-specific rough-weight input has the form

$$H(n) = (1 * g)(\mathbf{v}(n)) \mathbf{1}_{\{P^-(n) \geq n^\sigma\}}, \quad |H(n)| \ll_{g, \nu_0} 1. \quad (28)$$

Ford–Maynard’s Section 7 contains the corresponding rough-support mechanism; Lemma 7.18 is used in the proof of Theorem 7.3 to restrict the support by a least-prime-factor condition, while Theorem 8.2 combines the Section 7 machinery with a small exceptional region [FM24]. In the formal bank, however, the exact shifted-prime instantiation (28) is deliberately left uninhabited until its complete source dictionary is certified.

The Gate-0/2 development separately records the aggregate N_2 cell-sum repair: after choosing the largest prime as the final sieve variable, using a coarse geometric mesh and aggregating the cell errors, the desired N_2 upper bound is reduced to the two-linear-forms upper-sieve and prefix-volume/Mertens–PNT inputs. The finite compiler and the ε -uniformity bookkeeping are banked; the analytic source input is not promoted to a Lean theorem.

D.5 Gate 2 as a conditional endgame

Within the frozen Gate-0/2 research ledger, once the earlier Type-I/comparison inputs are fixed, the two load-bearing antecedents for the final shifted-prime endgame are:

- (i) the full Ford–Maynard Type-II estimate at operating width $1/6$, with the universal divisor-bounded coefficient quantifiers required by the published Type-II predicate;
- (ii) the source-exact rough-weight/ N_2 input represented by (28) and its uniform cell-sum consequence.

We denote these schematically by

$$\text{FullFMTypeII}_{1/6} \quad \text{and} \quad \text{FMRoughN2}.$$

The deterministic endgame compiler then gives the programme-level conditional implication

$$\begin{aligned} & \text{FullFMTypeII}_{1/6} + \text{FMRoughN2} + \text{Gate-0 comparison inputs} \\ & \implies \text{positive shifted-prime/twin mass at scale } x. \end{aligned} \quad (29)$$

The Lean development verifies the logical dependence and, in particular, contains counterguards showing that Full Type II alone is not the complete Gate-2 antecedent. The analytic antecedents of (29) are not presently inhabited in Lean.

Theorem D.2 (Conditional endgame). *Assume the frozen Gate-0 comparison/Type-I inputs, the full Ford–Maynard Type-II estimate on the interval $[\varepsilon, 1/6 - \varepsilon]$, and the source-exact rough N_2 input described above. Then the frozen Gate-2 compiler yields positive shifted-prime/twin mass at the scale x . If these hypotheses hold along arbitrarily large scales, the corresponding finite positivity implication produces infinitely many twin-prime pairs.*

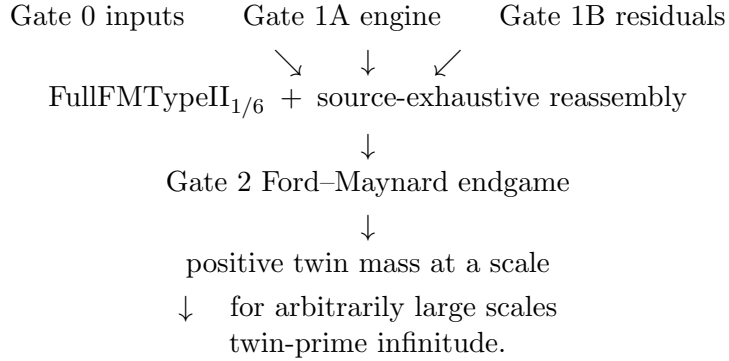
Proof. This is the deterministic dependency implication encoded by the Gate-0/2 endgame package: the hypotheses supply the two analytic antecedents missing from the finite compiler, while the earlier Gate-0 comparison inputs are held fixed. Positive twin mass at a scale has a finite support witness, hence produces a twin-prime pair at that scale. Repeating this on arbitrarily large scales gives infinitely many distinct pairs. No assertion is made here that the analytic hypotheses have been proved by the present manuscript. $\square$

If the package in (29) is available at arbitrarily large scales, the finite positivity compiler yields twin-prime pairs at arbitrarily large scales, hence infinitude. This is a *conditional logical consequence*, not a theorem proved by the current manuscript.

D.6 Where Gate 1A and Gate 1B enter Full Type II

The Ford–Maynard Type-II hypothesis is stronger than a collection of source-specific estimates: it quantifies over arbitrary divisor-bounded coefficient sequences on the full interval (25). Consequently neither Gate 1A alone nor the pair “Gate 1A + currently banked Gate 1B fragments” automatically supplies FullFMTTypeII_{1/6}. A separate source-exhaustive reassembly theorem is required.

The present programme therefore has the following honest dependency graph:



At present, the Gate-1B arbitrary-log residuals and the Full-Type-II reassembly remain open, so the final two arrows are not activated.

D.7 Publication-facing Gate-0/2 status

For clarity, the downstream status is summarized as follows:

Object	Current status
Gate-0 deterministic Type-I compiler	Lean-banked finite compiler
Gate-0 deep analytic inputs	research pass / external-uninhabited
$1/6 > 0.1663$ and margin $11/30000$	Lean-banked arithmetic
$\sigma \geq 49/300$	Lean-banked arithmetic
$\Omega(n) \leq 6$ for σ -rough n	Lean-banked theorem
Ford–Maynard rough-weight source dictionary	external/source pin; not a Lean inhabitant
N_2 cell-sum and ε -uniform splice	conditional research pass; Lean-uninhabited
Full FM Type II at width $1/6$	OPEN / uninhabited
Gate 2	CLOSED ONLY CONDITIONALLY on the displayed antecedents
Full Type-II reassembly	OPEN
Twin-prime infinitude	NOT PROVED

Status note D.3 (Full-programme closure status). The role of this section is to make the strongest currently defensible conditional closure explicit. It does not upgrade any external or source-inspected input to a theorem. The mathematical frontier remains the source-exhaustive Full-Type-II reassembly, with Gate 1B supplying the largest currently open analytic pieces.

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